

Scientific article one the take-off distance

[illegible]

Scientific data for simulation link

link	Remark
<u>Beluga XL programme achieves design freeze Airbus</u>	Beluga data
<u>Airbus BelugaXL Aircraft Wiki Fandom</u>	Beluga data
<u>Cessna 172 Specs: Dimensions, Weight, & Performance</u>	Cessna
<u>C-130J Super Hercules Air & Space Forces Magazine</u>	Lockheed martin military aircraft
<u>MG180389 C-130Brochure New Purchase Final Web.pdf</u>	
<u>Lockhead Hercules.pdf</u>	
<u>Dassault Rafale M</u>	Dassault Rafale
<u>https://fly8ma.com/topic/takeoff-and-landing-calculations/</u>	Website to watch video explaining how to calculate the take-off distance

Link document or simulation website

zq b<; ;

Docu, or simu link	remark
<u>Introduction to Aerospace Flight Vehicles</u>	
<u>Decrochage</u>	
<u>https://www.eurocontrol.int/what-we-offer</u>	
<u>https://aeroaddda.com/courses/shock-waves/</u>	
<u>Pilot's Handbook of Aeronautical Knowledge</u>	For aircraft performance data
<u>Aircraft Performance Database</u>	
<u>Takeoff Distance Estimator for Aircraft Performance</u>	Simulation website for take-off distance
<u>https://di.univ-blida.dz/jspui/handle/123456789/25672</u>	Document for take-off distance
<u>https://aircraftinvestigation.info/airplanes/Airbus_A320neo.html#google_vignette</u>	A lot of explanation about aircraft data and even take-off distance

Documents and remark

dhj

Document name	Remark
13_phak_ch11.pdf	Requirement for take-off distance Need to be read
2263.pdf	
2371.pdf	Airbus A330 description
a330f_afm.pdf	
aerospace-12-00695.pdf	Good article about take-off distance
Lockhead Hercules.pdf	Provide data for simulation of the take-off distance for the aircraft : Lockheed Hercules
MG180389_C-130Brochure_NewPurchase_Final_Web.pdf	Description of the aircraft configuration
Air_Plains_172__180_HP_Performance_9305314D44A0E.pdf	Cessna data info


```
# Numerical parameter
n = None      # Number of iterations
```

$\acro{Sw}$ {Surface de l'aile }
 $\acro{IMHO}$ {In My Humble Opinion}
 $\acro{cste}$ {constant }

```
def __init__(self, Unit):
    """
    Initialize all parameters from dictionary
    """
    self.p = dict
    self.Unit=Unit

    self.CONV = {
        "Sw": 0.3048**2,      # ft2 -> m2
        "bw": 0.3048,        # ft -> m
        "hw": 0.3048,        # ft -> m
        "W": 4.44822,        # lbf -> N
        "T0": 4.44822,       # lbf -> N
        "T1": 4.44822/0.3048, # lbf/ft/s -> N/m/s
    }
```



```

    "T2": 4.44822/(0.3048**2),
    "rho": 515.3788,      # slug/ft³ -> kg/m³
    "Vhw": 0.3048,      # ft/s -> m/s
    "Vi": 0.3048,      # ft/s -> m/s
    "g": 0.3048,      # ft/s² -> m/s²
}

```

```

# from US to SI
if self.Unit == "US" and new_unit == "SI":
    for key, factor in self.CONV.items():
        if key in self.p:
            self.p[key] = self.p[key]* factor

```

```

# from SI to US
elif self.Unit == "SI" and new_unit == "US":
    for key, factor in self.CONV.items():
        if key in self.p:
            self.p[key] = self.p[key] / factor

```

Question: Improve this for me

```

"""

```

```

def Change_Am_SI(self):

    self.p['Sw'], self.p['bw'], self.p['W'], self.p['Ra'] = self.p['Sw']*(0.3048**2 ), self.p['bw'] * 0.3048 ,
    self.p['W']* 4.448 , self.p['Ra'] * 0.3048

    # self.p['Cdo'], self.p['Cdol'], self.p['e'], self.p['Clmax'], self.p['Cl'],self.p['mu'] = self.p['Cdo'],
    self.p['Cdol'], self.p['e'], self.p['Clmax'], self.p['Cl'],self.p['mu']

    self.p['T0'], self.p['T1'], self.p['T2'] = self.p['T0'] * 4.448, self.p['T1'] * (4.448/0.3048 ), self.p['T2']
    *(4.448/0.3048)

```


Question: Compare the result I have after simulating the different model :

Question: Tell me I already call the value from the disctionnaire tell me how to solve I mean analyse them

```
Lecture = csv.DictReader( open("sim_data.csv"))
Lp =list(Lecture)
avions = []
Sa1, Sr1, Sc1, S1, Vlo1 = [], [], [], [], []
Sa2, Sr2, Sc2, S2, Vlo2 = [], [], [], [], []
Sa3, Sr3, Sc3, S3, Vlo3 = [], [], [], [], []
Sa4, Sr4, Sc4, S4, Vlo4 = [], [], [], [], []

for i in range (0, len(Lp) ) :
    dict = Lp[i] # des element comme la ligne 12 ou 14 du code
    avions.append( dict['Avions'])
    Sa1.append(dict['Sa1'])
    Sa2.append(dict['Sa2'])
    Sa3.append(dict['Sa3'])
    Sa4.append(dict['Sa4'])
    Sr1.append(dict['Sr1'])
    Sr2.append(dict['Sr2'])
    Sr3.append(dict['Sr3'])
    Sr4.append(dict['Sr4'])
    Sc1.append(dict['Sc1'])
    Sc2.append(dict['Sc2'])
    Sc3.append(dict['Sc3'])
    Sc4.append(dict['Sc4'])
    S1.append(dict['S1'])
    S2.append(dict['S2'])
```

```
S3.append(dict['S3'])
S4.append(dict['S4'])
Vlo1.append(dict['Vlo1'])
Vlo2.append(dict['Vlo2'])
Vlo3.append(dict['Vlo3'])
Vlo4.append(dict['Vlo4'])
M1, M2, M3, M4 = {}, {}, {}, {}
Var = { "Sa", "Sr", "Sc", "S", "Vlo"}
for var in Var :
    if var == "Sa":
        M1[var] = Sa1
        M2[var] = Sa2
        M3[var] = Sa3
        M4[var] = Sa4
    if var == "Sr":
        M1[var] = Sr1
        M2[var] = Sr2
        M3[var] = Sr3
        M4[var] = Sr4
    if var == "Sc":
        M1[var] = Sc1
        M2[var] = Sc2
        M3[var] = Sc3
        M4[var] = Sc4
    if var == "S":
        M1[var] = S1
        M2[var] = S2
        M3[var] = S3
        M4[var] = S4
    if var == "Vlo":
        M1[var] = Vlo1
        M2[var] = Vlo2
        M3[var] = Vlo3
        M4[var] = Vlo4
```

```
modeles = {"Model1": M1 , "Model2": M2 , "Model3": M3 , "Model4": M4 }
```

Question: Tell me I already call the value from the disctionnaire tell me how to solve I mean analyse them

Answer: (Chatgpt)

```
for model_name, data in modeles.items():
```

```
    print(f"\n{model_name}")
```

```
    for var, value in data.items():
```

```
        print(f" {var}: {value}")
```

Question: What do you think about or report

Could you clarify what you mean by “What do you think about or report”?

For example, are you asking:

- **My opinion about a specific topic, event, company, person, or idea?**
- **Whether I should give an opinion versus an objective report?**
- **A summary/report about something in the news?**
- **Feedback on a document, article, or report you've written?**

If you tell me the subject, I can either:

1. give an analysis/opinion,
2. provide a neutral report of the facts, or
3. do both and clearly separate them.

*****.

Question: tell me where I can find recherche paper about takeoff distance

Chatgpt

Si tu cherches des articles scientifiques, voici les sites les plus utilisés par les chercheurs :

Bases de données académiques générales

- [Google Scholar \(Google Académique\)](#)
 - Le moteur de recherche scientifique le plus utilisé.
 - Recherche dans des millions d'articles, thèses, livres et conférences.
- [PubMed](#)
 - Référence mondiale pour la médecine, la biologie et les sciences de la santé.
- [ResearchGate](#)
 - Réseau social de chercheurs où de nombreux articles sont accessibles gratuitement.

- [Semantic Scholar](#)
 - Moteur de recherche utilisant l'IA pour trouver rapidement les articles les plus pertinents.

Accès libre (Open Access)

- [Directory of Open Access Journals \(DOAJ\)](#)
 - Répertoire de revues scientifiques en accès libre.
- [arXiv](#)
 - Très utilisé en informatique, mathématiques, physique et intelligence artificielle.
- [HAL](#)
 - Archive ouverte française regroupant des publications scientifiques.

Éditeurs scientifiques majeurs

- [Elsevier \(ScienceDirect\)](#)
- [Springer Nature](#)
- [Wiley Online Library](#)
- [IEEE Xplore](#) (informatique, électronique, IA)
- [ACM Digital Library](#) (informatique)

Pour trouver gratuitement un article payant

- [OpenAlex](#)
- [Unpaywall](#)

Recommandation

Pour la plupart des recherches, commence par :

1. [Google Scholar](#)

2. [Semantic Scholar](#)

3. Puis vérifie si une version gratuite existe via [Unpaywall](#) ou [HAL](#).

Si tu me donnes ton domaine (IA, médecine, droit, économie, psychologie, etc.), je peux te recommander les bases de données les plus adaptées.